
Inverting Retargeting: Humanoid Datasets Remember Their Operators

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Abstract

1 Human-to-humanoid motion retargeting is the central operation in teleoperation
2 data collection: it projects operator demonstrations onto a shared robot skeleton,
3 discarding body shape, to produce training data for whole-body controllers and
4 foundation models. We report a surprising property of this transform: **retargeting
5 normalizes body proportions but preserves operator movement dynamics**
6 (joint velocity profiles, ranges of motion, and coordination patterns shaped by the
7 operator’s physiology). On BONES-SEED (522 operators, 142K sequences), retar-
8 geted Unitree G1 trajectories support gender classification at 96.0% and operator
9 re-identification at 97.2% Top-1; on *operators never seen during training*, gender
10 holds at 83.4% and age and height regress within ± 4.2 yr and ± 5.7 cm. Partial
11 correlation analysis reveals emergent, biomechanically interpretable structure: the
12 signals are task-invariant across activity categories and hold across retargeting
13 implementations. We introduce UNVEIL, a skeleton-aware spatiotemporal graph
14 network to measure and interpret this effect, take initial steps toward operator
15 anonymization, and ask what this implies for data practices and the operator-
16 invariance of learned policies. Code, models, and anonymized trajectories are at
17 our project website <https://project-unveil.github.io>.

18 1 Introduction

19 Humanoid robot learning is scaling. Our community has built whole-body teleoperation systems
20 that transfer operator motion to humanoids in real time [7, 8, 11, 31, 2], shown that training motion-
21 tracking controllers on motion capture data improves consistently with data volume and compute [16,
22 31, 1], and is now assembling large shared datasets as training infrastructure for humanoid policies
23 and foundation models.

24 **Motion retargeting** is the operation at the center of this data pipeline. Because a human body and a
25 humanoid like the Unitree G1 differ in morphology and joint count, an operator’s motion cannot be
26 executed on the robot directly. Retargeting solves for the robot’s joint configuration at each frame by
27 minimizing positional error at a set of shared landmarks (head, hands, feet, torso), subject to joint
28 limits [12, 31, 1]. This discards body-shape parameters entirely: limb proportions, mass distribution,
29 and limb lengths are stripped away, and every operator’s motion is projected onto the same robot
30 joint space regardless of their physique (Figure 1). Crucially, this inverse-kinematics solve operates
31 frame-by-frame: it matches landmark positions at each instant but places no objective over how
32 joints move across time. Joint velocity profiles, inter-limb coordination rhythms, and movement
33 cadences shaped by the operator’s physiology and motor habits pass through the solve untouched.
34 The resulting G1 trajectory describes the robot’s motion, not the person who produced it; retargeting
35 was designed as a one-way transform. We show it is not entirely one-way. The retargeting pipeline is,
36 in a meaningful sense, *invertible*: **retargeting removes body morphology yet preserves operator
37 movement dynamics**, and those dynamics are recoverable from the robot-space data alone.

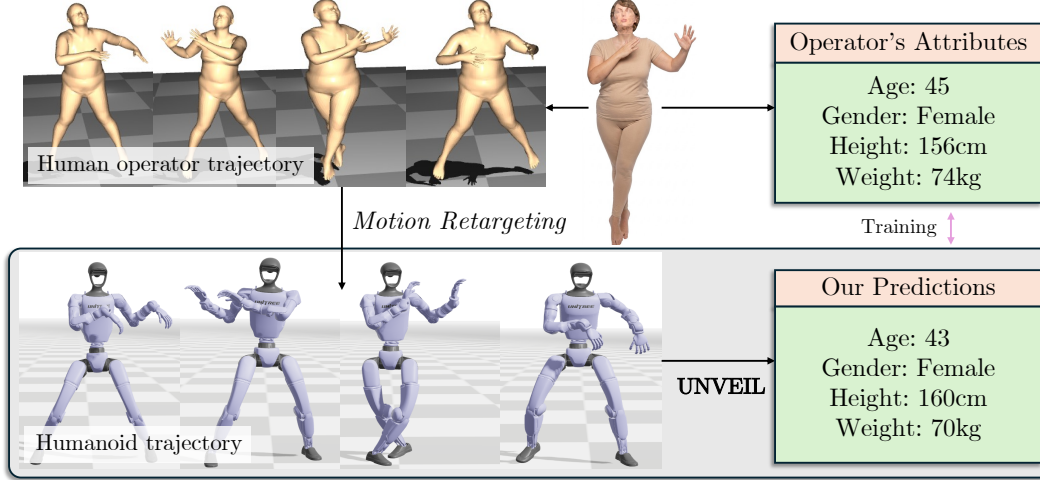


Figure 1: Operator movement signatures survive the retargeting transform. Retargeting projects diverse teleoperators onto a uniform robot skeleton (Unitree G1), stripping body-shape parameters entirely. Operator-specific movement dynamics nevertheless persist in the joint-angle trajectories, and our proposed UNVEIL recovers the operator’s attributes directly from the humanoid data.

To measure what survives, we introduce UNVEIL, a skeleton-aware spatiotemporal GCN that encodes position, velocity, and acceleration per joint, models inter-joint dependencies through two hierarchies (intra-limb, limb-torso), and predicts operator attributes from retargeted G1 trajectories. In this sense, UNVEIL operationalizes a tractable and revealing form of *inversion*: recovering operator identity and physique attributes directly from retargeted trajectories, without access to the original human motion. We validate this on BONES-SEED [3]: 142,220 sequences from 522 operators across locomotion, manipulation, dance, and everyday activities, retargeted onto the Unitree G1 with per-operator demographic metadata. On held-out demonstrations from operators seen during training, UNVEIL achieves 96.0% gender classification and 97.2% Top-1 operator re-identification ($\sim 400\times$ above chance). On operators the model has *never encountered before*, gender classification holds at 83.4% (+33 pp above chance) and age, height, and weight regress within ± 4.2 yr, ± 5.7 cm, and ± 9.1 kg respectively.

The effect has emergent, biomechanically interpretable structure: joint velocity profiles, ranges of motion, and limb coordination patterns encoding the operator’s build survive retargeting and leave consistent traces in the robot trajectories. Partial correlation analysis (which isolates each attribute’s contribution after accounting for the others) confirms these are genuine independent signals: older operators produce less trunk rotation across every activity category; taller operators consistently cover less root distance per unit time; establishing the leakage as a structural property of the retargeting transformation rather than a dataset artifact.

As teleoperation datasets scale and retargeted trajectories become shared training infrastructure, understanding what they retain about the operators who produced them matters both scientifically and practically, for data sharing practices and for the operator-invariance of learned policies. We release code, model weights, anonymized trajectories, and result visualizations at our project website.

Our contributions are: (1) we identify and characterize operator movement signature leakage in retargeted humanoid trajectories, a previously unexamined property of the human-to-humanoid teleoperation pipeline, across six inference tasks and 522 operators (Section 4); (2) we introduce UNVEIL, a skeleton-aware spatiotemporal GCN that substantially exceeds random baselines on all tasks, with ablation and partial-correlation analysis grounding the leakage in interpretable, task-invariant motion features (Section 3); (3) we take steps toward operator anonymization and outline open problems in privacy-aware dataset curation and operator-invariant policy learning, establishing a structure and baseline for future work in our community (Section 4).

69 2 Related Work

70 **Humanoid teleoperation and retargeting.** Teleoperation systems that transfer full-body operator
 71 motion to humanoids in real time [8, 7, 2, 6] rely on retargeting pipelines that project human motion
 72 onto the robot’s joint space [12, 11, 31, 1]. SONIC [16] shows that controllers trained on such
 73 retargeted data scale favorably with data volume and compute, motivating large shared corpora of
 74 retargeted trajectories as humanoid training resources. BONES-SEED [3], the motion-capture corpus
 75 central to SONIC’s training pipeline, pairs retargeted G1 trajectories with per-operator demographic
 76 metadata, making operator attribute inference from retargeted data tractable at scale. That retargeting
 77 discards body-shape parameters is well understood; what it preserves about operator movement has
 78 not been studied. We study that question.

79 **Skeletal motion understanding.** Spatiotemporal GCNs are the standard architecture for motion
 80 understanding from joint sequences, from ST-GCN [30] and adaptive two-stream variants [24] to
 81 more recent work with learned multi-scale graph structures [15, 5, 10, 14, 29]. The same family has
 82 been applied to person re-identification from skeleton data [25, 17], shifting the objective from action
 83 classification to performer identification. Unlike prior work in this family, UNVEIL operates on
 84 robot-space joint angles rather than human-pose sequences, with its graph defined over the robot’s
 85 kinematic groups and its task being operator attribute recovery from retargeted data rather than action
 86 classification or performer re-identification. The robot’s kinematic tree differs from a human body
 87 skeleton in topology and the semantics of each degree of freedom, motivating a graph structure built
 88 from the robot’s morphology rather than adapted from a human-pose design. The three-stream input
 89 (position, velocity, acceleration) and intra/inter-limb branch design are specific to the retargeting
 90 setting: body shape is stripped but movement dynamics survive, and operator-specific information
 91 lives in these dynamics, not in body configuration.

92 **Operator movement signatures in human motion.** Human motion is biometrically rich: skeleton
 93 sequences leak re-identifiable identity and demographic attributes in action-recognition settings [17, 4,
 94 25], and in XR environments users can be identified from head and hand motion alone [22, 18, 28, 19].
 95 Retargeting is not merely a change of representation: it clips kinematics to the robot’s feasible joint
 96 range and maps every operator’s motion to the same joint space regardless of physique, making
 97 the persistence of individual signatures a non-trivial finding. Orthogonal to our focus, prior work
 98 has responded with anonymization methods operating on human-space motion directly: adversarial
 99 perturbation of joint sequences [17] and retargeting onto dummy skeletons [27]. Whether these
 100 signatures survive retargeting onto a robot morphology has not been studied. We show that operator
 101 movement signatures do survive retargeting, and that the leakage is structural: it holds across activity
 102 types, persists for operators never seen during training, and reflects interpretable patterns in joint
 103 velocity and coordination timing.

104 **Toward operator-invariant robot learning.** Complementing scaling efforts that have assembled
 105 large retargeted datasets for humanoid learning [20, 9, 16], our work raises a question those efforts
 106 leave open: whether demonstrator characteristics propagate into trained policies, with implications
 107 for generalization and responsible data sharing [23]. If policies trained on retargeted demonstrations
 108 learn to rely on operator-specific motion cues, their behavior may not transfer cleanly to new operators
 109 or to deployment without a fixed human partner. We provide an initial characterization of this leakage
 110 and a baseline for future work on operator-invariant policy learning and data curation.

111 3 UNVEIL

112 We propose UNVEIL, a spatiotemporal network that takes a retargeted humanoid motion trajectory
 113 as input and predicts operator attributes, including age, gender, weight, height, and identity. As
 114 illustrated in Figure 2, UNVEIL first encodes the raw trajectory into per-joint motion features,
 115 such as velocity and acceleration, through a kinematic encoder (Section 3.1). Next, it captures
 116 joint coordination patterns at two hierarchies (intra-limb and limb-torso) through a spatial graph
 117 convolution (Section 3.2). The resulting per-frame features are then aggregated across time by a
 118 temporal convolution (Section 3.3), and pooled into a single embedding (Section 3.4).

119 **Preliminaries and Notation.** We model the humanoid as a kinematic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where
 120 $\mathcal{V}$ is the set of $J = |\mathcal{V}|$ joints and $\mathcal{E}$ is the set of edges connecting every pair of joints that are
 121 mechanically linked by a single rigid body segment. A humanoid trajectory is a sequence of frames

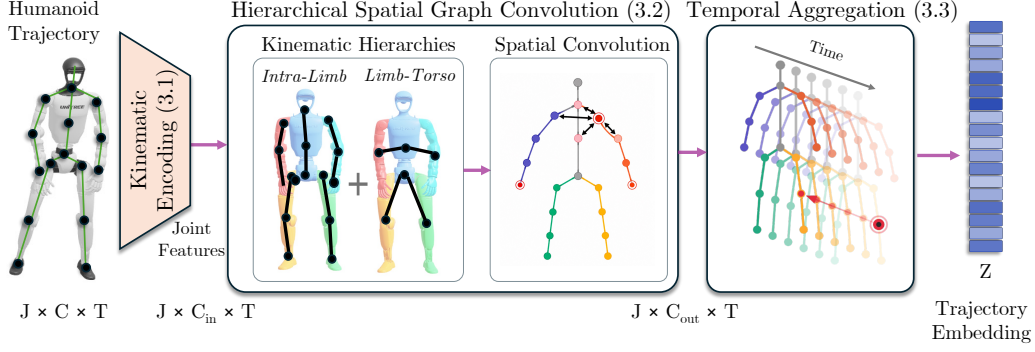


Figure 2: **UNVEIL overview.** A retargeted humanoid trajectory is encoded into per-joint motion features, processed by two hierarchical spatial graph convolutions (intra-limb, limb-torso), aggregated across time with N spatiotemporal layers into a trajectory embedding that is decoded by task-specific heads into operator attributes.

$\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_T)$, with $\mathbf{X}_t \in \mathbb{R}^{J \times C}$ the joint state at frame t and C the per-joint state dimension. The state of joint j at frame t is $\mathbf{x}_{j,t} \in \mathbb{R}^C$, representing each joint’s degrees of freedom (DoF) (e.g., a one-dimensional rotation angle suffices for the elbow, whereas the shoulder requires three rotation angles). We partition the joints into five disjoint limb sets, $\mathcal{V} = \mathcal{V}_{tr} \cup \mathcal{V}_{La} \cup \mathcal{V}_{Ra} \cup \mathcal{V}_{Ll} \cup \mathcal{V}_{Rl}$, corresponding to the torso, left arm, right arm, left leg, and right leg.

3.1 Kinematic Encoding

We first transform the raw trajectory to capture the motion dynamics of each joint. Our key insight is that operator attributes are reflected in *how* joints move rather than in where they are. Consider two operators of different heights teleoperating the humanoid to reach the same point on a table: their final joint state may be nearly identical, yet the shorter operator must extend the arm more aggressively, producing a sharper acceleration at the elbow joint than the taller operator, whose reach is comfortably within range. We therefore encode the trajectory with a temporal convolution applied independently per joint, mapping the raw trajectory $\mathbf{X} \in \mathbb{R}^{J \times C \times T}$ to a joint feature tensor $\mathbf{H} \in \mathbb{R}^{J \times C_{in} \times T}$, capturing motion characteristics such as velocity and acceleration.

3.2 Hierarchical Spatial Graph Convolution

$\mathbf{H}$ summarizes each joint independently and does not yet model joint interactions. Traditional skeleton-based GCNs capture these via a single adjacency matrix [30], but operator attributes are reflected in coordination patterns at multiple spatial scales that a single adjacency cannot separate: elbow-wrist coordination during a short reach is a fine-grained, intra-limb signal, while torso-arm coupling during a long reach is a coarser signal, both informative of operator height. We therefore decompose $\mathcal{G}$ into two hierarchies, each capturing joint interactions at a distinct spatial scale.

Intra-limb Interactions. Fine-grained motions (e.g., waving) are characterized by the coordination between joints within a single limb, for instance, between the elbow and the wrist of one arm. We capture these interactions with the intra-limb hierarchy $\mathcal{H}_1 = \{\mathcal{G}_k^{(1)}\}_{k=1}^5$, which contains one subgraph per limb. Each subgraph $\mathcal{G}_k^{(1)} = (\mathcal{V}_k, \mathcal{E}_k)$ retains only the joints and edges internal to limb k , such that a spatial graph convolution applied to $\mathcal{G}_k^{(1)}$ propagates features within that limb.

Limb-torso Interactions. Larger motions, such as reaching forward to grasp an object, require the limb and the torso to move together, since the torso must rotate or lean to extend the limb’s effective range. We capture these interactions with the limb-torso hierarchy $\mathcal{H}_2 = \{\mathcal{G}_k^{(2)}\}_{k=1}^4$, with each subgraph corresponding to a distinct extremity limb $m \in \{La, Ra, Ll, Rl\}$. For limb k , the corresponding subgraph $\mathcal{G}_k^{(2)} = (\mathcal{V}_k \cup \mathcal{V}_{tr}, \mathcal{E}_k \cup \mathcal{E}_{tr} \cup \{e_k\})$ pairs that limb with the torso through the connecting edge e_k , such that a spatial graph convolution applied to $\mathcal{G}_k^{(2)}$ propagates features between the limb and the torso while remaining isolated from the other limbs.

Adjacency Representation. We represent each subgraph $\mathcal{G}_k^{(h)}$ across the two hierarchies by an adjacency matrix $\mathbf{A}_k^{(h)} \in \{0, 1\}^{J \times J}$ defined over the complete joint set, with $\mathbf{A}_k^{(h)}[i, j] = 1$ iff $i, j \in \mathcal{V}_k^{(h)}$ and either $(i, j) \in \mathcal{E}_k^{(h)}$ or $i = j$. We use the symmetrically normalized form $\tilde{\mathbf{A}}_k^{(h)} = (\mathbf{D}_k^{(h)})^{-1/2} \mathbf{A}_k^{(h)} (\mathbf{D}_k^{(h)})^{-1/2}$, where $\mathbf{D}_k^{(h)}$ is the diagonal degree matrix of $\mathbf{A}_k^{(h)}$. This uniform shape allows us to combine intra-limb and limb-torso interactions in a single layer, so the network can learn the relative importance of fine-grained and compound motion patterns end-to-end.

Graph Convolution Layer. We denote the joint features at frame t by $\mathbf{H}_t \in \mathbb{R}^{J \times C_{\text{in}}}$. We assign each subgraph $\mathcal{G}_k^{(h)}$ its own learnable weight matrix $\mathbf{W}_k^{(h)} \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}}}$ that maps each joint’s C_{in} -dimensional input feature to a C_{out} -dimensional output feature, such that the same joint feature is transformed differently depending on which subgraph it is being aggregated over. We index the subgraphs of hierarchy h by $\mathcal{K}_h$, with $|\mathcal{K}_1| = 5$ and $|\mathcal{K}_2| = 4$, and define the hierarchical spatial graph convolution operator $\mathcal{S}$ frame-by-frame as

$$\mathcal{S}(\mathbf{H}_t) = \sigma \left(\sum_{h=1}^2 \sum_{k \in \mathcal{K}_h} \tilde{\mathbf{A}}_k^{(h)} \mathbf{H}_t \mathbf{W}_k^{(h)} \right). \quad (1)$$

Equation (1) aggregates each joint’s feature with the features of its neighbors in each subgraph, then sums these aggregations across the two hierarchies. The summation is a learned, weighted combination (the magnitude of $\mathbf{W}_k^{(h)}$ controls how much each hierarchy contributes), allowing the network to adapt the relative importance of the two scales to the attribute being predicted.

3.3 Temporal Aggregation

The spatial graph convolution (Equation (1)) treats each frame in isolation. To capture how joint features evolve over time, we first apply $\mathcal{S}$ to each frame independently, using shared weights across all T frames, to obtain a per-frame representation $\mathbf{Y} \in \mathbb{R}^{J \times C_{\text{out}} \times T}$. We then aggregate information across frames with a one-dimensional temporal convolution operator $\mathcal{T}$ applied per joint:

$$\mathcal{T}(\mathbf{Y})_{j, :, t} = \sum_{\tau = -\lfloor \Gamma/2 \rfloor}^{\lfloor \Gamma/2 \rfloor} \mathbf{W}_{:, :, \tau}^{\text{tmp}} \mathbf{Y}_{j, :, t+\tau}, \quad (2)$$

where Γ is the kernel size and $\mathbf{W}^{\text{tmp}} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{out}} \times \Gamma}$ is the learned temporal kernel. Concretely, for joint j and frame t , $\mathcal{T}$ collects the per-joint feature vectors $\mathbf{Y}_{j, :, t+\tau} \in \mathbb{R}^{C_{\text{out}}}$ across a Γ -frame window around t , applies the corresponding kernel slice $\mathbf{W}_{:, :, \tau}^{\text{tmp}} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{out}}}$ to each, and sums the results into a single C_{out} -vector. The kernel has no joint index, so $\mathcal{T}$ is a standard 1D convolution along time applied with the same weights to every joint’s features. Applying $\mathcal{T}$ per joint reflects that the operator-specific motion pattern at a given joint is a property of that joint’s own movement; cross-joint interactions are already handled by the spatial convolution that precedes it. A single application spans Γ frames. Stacking L spatiotemporal layers extends the receptive field to $\mathcal{O}(L \cdot \Gamma)$ frames, with strided convolutions at selected depths further extending coverage while reducing temporal resolution (Appendix B).

3.4 Network Architecture and Training

We compose the three components into a single architecture by stacking L spatiotemporal layers, each composing $\mathcal{S}$ (Equation (1)) and $\mathcal{T}$ (Equation (2)) with batch normalization and a residual connection. We compute layer l ’s output $\mathbf{H}^{(l)}$ recursively, starting from the kinematic encoding $\mathbf{H}^{(0)} = \mathbf{H}$ and applying $\mathbf{H}^{(l+1)} = \mathcal{T}(\mathcal{S}(\mathbf{H}^{(l)})) + \mathbf{H}^{(l)}$ for $l = 0, \dots, L-1$. The channel width increases with depth and strided temporal convolutions are applied at a small number of selected layers (specified in Appendix B) to progressively downsample the time axis, producing a final joint feature tensor $\mathbf{H}^{(L)} \in \mathbb{R}^{J \times D \times T_L}$ where $T_L < T$ is the downsampled temporal length and D is the embedding dimension. This schedule lets early layers capture short-range, fine-grained joint dynamics, while deeper layers integrate over longer temporal spans at coarser resolution. We pool $\mathbf{H}^{(L)}$ globally over both the joint and time dimensions to obtain

$$\mathbf{Z} = \frac{1}{J \cdot T_L} \sum_{j=1}^J \sum_{t=1}^{T_L} \mathbf{H}_{j, :, t}^{(L)}, \quad (3)$$

where $\mathbf{H}_{j,:t}^{(L)} \in \mathbb{R}^D$ denotes the slice of $\mathbf{H}^{(L)}$ at joint j and frame t and $\mathbf{Z} \in \mathbb{R}^D$ is the humanoid trajectory embedding.

Training. We train a separate UNVEIL instance per attribute. Classification tasks (gender, re-id) minimize cross-entropy with label smoothing; regression tasks (age, height, weight) minimize MSE. For re-identification, we add a supervised contrastive [13] auxiliary loss that pulls same-operator embeddings together and pushes different operators apart, with a warmup phase before the contrastive term activates. The same embedding space supports linkage: two trajectories are deemed same-operator if their cosine similarity exceeds a validation-tuned threshold. Full hyperparameters and architecture details are in Appendix B.

3.5 Operator-Aware Anonymization

As an initial step toward suppressing the leakage we have characterized, we design an operator-aware anonymization approach. Our anonymizer disentangles a trajectory into a motion factor and a privacy factor, then substitutes the privacy factor with that of a target whose operator differs from the source’s, producing a trajectory that no longer reveals the source operator’s attributes.

Anonymizer. We use a motion encoder E_M that captures the action content, a privacy encoder E_P that captures the operator’s attribute signature, and a shared decoder D that reconstructs a trajectory from the two embeddings. At inference, we anonymize a source trajectory by replacing its privacy embedding with that of a target trajectory $\mathbf{X}_{\text{target}}$, yielding a trajectory, $\tilde{\mathbf{X}} = D(E_M(\mathbf{X}), E_P(\mathbf{X}_{\text{target}}))$, that preserves the source’s action while carrying the target’s attribute signature.

Training and Metrics. We train $\{E_M, E_P, D\}$ jointly against a panel of attribute classifiers $\{f_a\}_a$, each an instance of UNVEIL trained to predict an operator attribute a . The encoders minimize a reconstruction loss together with an adversarial loss against the classifiers,

$$\mathcal{L} = \|D(E_M(\mathbf{X}), E_P(\mathbf{X})) - \mathbf{X}\|_2^2 + \lambda \sum_a [\mathcal{L}_{\text{cls}}(f_a(E_P(\mathbf{X})), y_a) - \mathcal{L}_{\text{cls}}(f_a(E_M(\mathbf{X})), y_a)], \quad (4)$$

while the classifiers $\{f_a\}$ are alternately updated to minimize their loss on both encoder outputs. The adversarial term concentrates each attribute signature in E_P by rewarding the classifiers’ recovery from $E_P(\mathbf{X})$ and penalizing it from $E_M(\mathbf{X})$, so that swapping E_P perturbs the attribute signature while leaving the action representation in $E_M(\mathbf{X})$ intact. For each source $\mathbf{X}$, we set $\mathbf{X}_{\text{target}}$ to the training trajectory whose operator is most distant from $\mathbf{X}$ ’s in standardized (age, gender, weight, height) space, yielding a sample-specific contrast.

We measure privacy by the attribute-recovery accuracy of UNVEIL and motion utility by an action classifier accuracy trained on the BONES-SEED activity labels on $\tilde{\mathbf{X}}$.

4 Experiments

In this section, we include UNVEIL’s evaluation setup (Section 4.1), present quantitative results across the six inference tasks (Section 4.2), and report ablations and interpretability analyses that characterize the emergence of operator-specific motion patterns in the retargeted trajectories (Section 4.3).

4.1 Setup

BONES-SEED Dataset [3]. A large-scale motion-capture dataset from [16] containing 142,220 G1-retargeted sequences from 522 operators across 20 activity categories (locomotion, dance, sports, everyday activities, etc.). Operators span ages 17–71 yr, heights 145–199 cm, weights 38–145 kg, with a balanced gender split. Each demonstration is available as both a human pose sequence and a G1 joint-angle trajectory, tagged with per-operator demographic metadata—making it uniquely suited for studying whether retargeted humanoid datasets remember their operators.

Two axes of generalization. We partition the operators into a training and held-out unseen sets.

- The *seen-operators* protocol evaluates generalization on unseen demos from operators present during training, measuring generalization across motion samples.
- The *unseen-operators* protocol evaluates on demos from operators never observed during training, measuring generalization across the operator population.

Table 1: UNVEIL’s performance across inference tasks and evaluation settings.

Task → Eval. Setting ↓	Gender Acc (%)↑	Weight MAE↓	Height MAE↓	Age MAE↓	Re-ID Top-1/5 Acc (%)↑	Linkage Acc (%)↑
Unseen operators	83.4	9.1 kg	5.7 cm	4.2 yr	—	—
Unseen demos of seen operators	96.0	4.0 kg	3.6 cm	2.5 yr	97.2/98.8	92.4
<i>Performance range across 20 activity categories</i>						
Most leaking activity	100.0	4.7 kg	3.3 cm	0.4 yr	100.0/100.0	100.0
2nd-most leaking	94.1	5.3 kg	3.9 cm	1.5 yr	98.0/99.0	99.0
2nd-least leaking	74.4	11.1 kg	7.4 cm	3.8 yr	70.1/89.5	74.9
Least leaking activity	72.9	14.8 kg	7.8 cm	3.9 yr	73.0/84.2	69.5

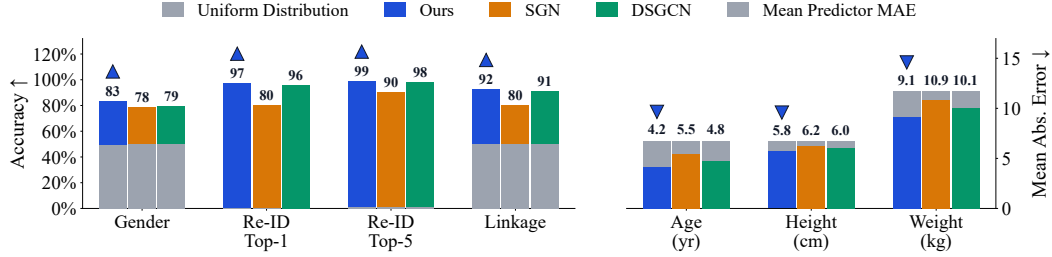


Figure 3: Comparison between UNVEIL and two well-tested action-recognition architectures adapted with our UNVEIL framework (SGN [32] and DSGCN [29]). We consistently improve over both adaptations across all tasks. Gray shows task-specific chance performance for reference.

Tasks and metrics. We evaluate UNVEIL on six operator attribute inference tasks from G1 trajectories. Gender classification reports accuracy. Age, height, and weight are continuous regression tasks evaluated by mean absolute error (MAE). Re-identification matches a query trajectory against a per-operator gallery of held-out demos via cosine similarity in the learned embedding space, reporting Top-1 and Top-5 accuracy. Linkage decides whether two trajectories were produced by the same operator by thresholding their pairwise cosine similarity, reporting accuracy. Re-id and linkage operate only under the seen-operators protocol.

Calibrating a first-of-its-kind result. This work provides the first evidence of operator attributes leaking through retargeted humanoid trajectories collected for robot learning. While no direct baselines exist, we calibrate our results in two ways. First, we report intuitive task-specific chance references: 50% accuracy for gender and linkage, the MAE of the mean predictor for regression tasks, and $1/N_{\text{operators}}$ for re-identification. Second, we instantiate UNVEIL with two well-tested action-recognition techniques, Semantic Conditioning [32] and Body-Part Typing [29] adapting their spatial graph designs into our pipeline. We state this up front for fair evaluation, include results and implementation details in Appendix A, and adapt code from the official releases.

4.2 Quantitative Results

Humanoid datasets do carry physical attributes of their operators (Table 1, Figure 3). From humanoid trajectories, UNVEIL can infer gender at 83.4% accuracy on operators never seen during training (+33.4% from chance). For continuous attributes under the same unseen-operators protocol, UNVEIL recovers age within ± 4.2 yr, height within ± 5.7 cm, and weight within ± 9.1 kg of the ground truth. On seen operators with held-out demonstrations, UNVEIL’s accuracy increases to 96.0% on predicting gender and regression errors decrease further; the gap confirms that the leakage generalizes across the operator population.

Retargeted trajectories function as soft biometrics for operator identity (Table 1). UNVEIL re-identifies the operator behind a retargeted G1 trajectory with 97.2% accuracy (Top-1), roughly $400\times$ the $1/N_{\text{operators}}$ random-chance baseline of 0.2%. The same embedding supports operator

Table 2: Ablation study of UNVEIL. Each row removes one component (marked ✗) while keeping the two others (marked ✓). ↓ indicates the change from the full model. Gray is our full model.

Components			Gender Accuracy(%)		Re-ID Accuracy(%)	
Kinematic Encoding	Skeleton-Aware Modeling	Temporal Aggregation	Seen Operator Unseen Demo	Unseen Operator	Top-1	Top-5
✓	✓	✓	96.0	83.4	97.2	98.8
✗	✓	✓	94.1 (↓1.9)	81.2 (↓2.2)	93.0 (↓4.0)	98.1 (↓0.7)
✓	✗	✓	93.9 (↓2.1)	81.3 (↓2.1)	94.7 (↓2.5)	98.0 (↓0.8)
✓	✓	✗	87.7 (↓8.3)	76.9 (↓6.5)	86.7 (↓10.5)	94.7 (↓4.1)

linkage (deciding whether two trajectories belong to the same operator) enabling 92.4% linkage accuracy against the 50% random-chance baseline. The near-saturation of aggregate performance indicates that retargeted humanoid trajectories carry operator-specific signatures strong enough to serve as soft biometrics, even after retargeting has discarded body-shape parameters.

Leakage persists across different activities (Table 1, bottom block). Per-activity performance varies but never collapses: motion-rich activities (jogging, dance) yield near-perfect accuracy, while the least indicative (crouching, eating) still achieve 72.9% gender accuracy (+22.9% over chance) and 73.0% Top-1 re-id. Privacy-relevant signal persists across activities rather than being concentrated in a handful of rich motions.

Findings generalize across modeling choices (Table 5, Appendix A). To verify that our findings are not contingent on a particular kinematic modeling strategy, we adapt two well-tested action-recognition architectures, SGN [32] and DSGCN [29] into our pipeline by replacing UNVEIL’s spatial graph with theirs while retaining our joint kinematic encoding and temporal aggregation. Both adaptations recover operator attributes well above chance across all tasks and evaluation protocols.

4.3 Ablations and Analyses

We evaluate each core component of UNVEIL by removing one at a time while keeping the other two intact, retraining from scratch on the same splits (Table 2).

Kinematic encoding captures motion dynamics beyond static pose. We bypass the kinematic encoder and feed raw joint angles directly. This costs ~2 pp on gender and 4 pp on Top-1 re-identification, confirming that learned motion features help isolate operator-specific dynamics.

Skeleton-aware modeling separates coordination at distinct spatial scales. We replace the hierarchical intra-limb and limb-torso branches with a single fully-connected graph. Gender accuracy drops by ~2 pp and Top-1 re-identification by 2.5 pp, indicating that decomposing the kinematic graph into meaningful substructures helps the network isolate operator-specific coordination patterns.

Temporal aggregation is the most critical component. We collapse each demonstration to a single mean frame, eliminating all temporal modeling. This produces the largest drops: gender accuracy falls by 8.3 pp on seen and 6.5 pp on unseen operators, and Top-1 re-identification decreases by 10.5 pp, consistent with UNVEIL’s premise that operator signatures live in temporal dynamics.

Interpreting why humanoid datasets remember their operators. Human-to-humanoid retargeting strips away body shape, but it does not fully strip away how an operator moves: velocity profiles, ranges of motion, and coordination timing are shaped by the operator’s physiology and persist in the retargeted trajectory. To quantify this, for a set of humanoid trajectories, we extract 27 interpretable features: for each of nine body regions (root, hip, knee, ankle, waist, shoulder, elbow, wrist, and root rotation), we compute three motion statistics, mean velocity, peak velocity, and range of motion and compute both raw Pearson and partial correlations to our four attributes (gender, weight, height, & age)¹. We highlight several interpretable patterns (Table 7):

- *Older operators move their torso less.* Waist range of motion correlates with age ($r=-0.30$), and the partial correlation holds at -0.25 after controlling for height, weight, and gender, confirming the

¹Partial correlations are necessary because these attributes are mutually correlated in human populations. So a feature that appears to track one attribute may in fact be borrowing signal from another (details deferred to Appendix C)

Table 3: Mean raw and partial correlations for the most strongly correlated features per attribute. Detailed table is provided in Appendix C (Table 7)

Attribute	Feature	Mean r	Mean r_{partial}
Age	waist_rom	−0.30	−0.25
	hip_peak_vel	−0.29	−0.28
Height	root_translate_rom	−0.25	−0.25
	root_translate_peak_vel	−0.19	−0.18
Weight	root_translate_peak_vel	−0.25	−0.20
	waist_rom	−0.20	−0.14
Gender	ankle_rom	−0.31	−0.18
	wrist_rom	−0.14	−0.13

Table 4: Evaluation of our anonymization. UNVEIL trained on original trajectories evaluates anonymized outputs. Operator Attributes: ↓ better for classification (Acc), ↑ better for regression (MAE). Utility: ↑ better for action-recognition accuracy.

	Operator Attributes						Utility	
	Re-ID	Linkage	Gender	Age	Height	Weight	Action Recognition	
	Top-1↓	Acc↓	Acc↓	MAE↑	MAE↑	MAE↑	Top-1↑	Top-5↑
Original	97.2%	92.4%	83.4%	4.2 yr	5.7 cm	9.1 kg	88.7%	98.3%
Anonymized	16.8%	53.1%	51.8%	7.6 yr	8.7 cm	10.7 kg	80.0%	98.1%

signal is age-specific, likely reflecting reduced spinal mobility.

- *Taller operators cover less root distance per unit time.* Root translational peak velocity correlates with height ($r=-0.19$), and the partial correlation (-0.18) remains nearly unchanged, indicating a genuinely height-specific relationship.
- *Heavier operators produce slower root motion.* Root peak velocity correlates with weight ($r=-0.25$), but the partial correlation drops to -0.20 , revealing that part of the raw signal was shared with other attributes, illustrating why partial correlations are necessary.

Anonymization sharply reduces operator-attribute leakage while preserving most motion utility (Table 4). Our anonymizer pushes re-identification from 97.2% to 16.8% Top-1, and the same trend holds consistently across gender and linkage, both collapsing to within a few points of their 50% chance floors. Regression MAEs roughly double, approaching mean-predictor errors. Motion utility is largely preserved: Top-1 action recognition drops by only ~ 9 pp while Top-5 stays effectively unchanged, confirming that operator-aware anonymization can mask movement signatures while retaining the motion content needed for downstream robot learning.

5 Conclusion

We have taken a first step toward characterizing and addressing operator privacy in retargeted humanoid datasets, and provide the structure necessary for advancing future work along several axes. The current analysis focuses on a single retargeting implementation and robot morphology (Unitree G1); our characterization could benefit from evaluation across retargeting pipelines and robot platforms to strengthen generality. Investigating whether operator signatures propagate into downstream policies trained on retargeted data remains an important direction that our framework is well-positioned to support. Our anonymization approach could be complemented by alternative architectures for motion disentanglement, such as variational autoencoders [21] or diffusion-based motion generation [26], to further improve the privacy–utility tradeoff in future work. We release code, model weights, and anonymized trajectories to support these directions.

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Appendix

- **Appendix A:** Additional results with two well-tested action-recognition architectures (SGN, DSGCN) adapted to our pipeline, following the same evaluation protocol as Table 1.
- **Appendix B:** Training details including hyperparameters, architecture specifics, and compute requirements.
- **Appendix C:** Correlation analysis details covering feature extraction, aggregation, task filtering, and the full raw and partial correlation table with sign retention.
- **Appendix D:** Dataset details including BONES-SEED attribute distributions and the Unitree G1 joint layout.

A Additional Results

We adapt two well-tested action-recognition architectures, SGN [32] (Semantic Conditioning) and DSGCN [29] (Body-Part Typing), to our pipeline by replacing UNVEIL’s spatial graph design with theirs while keeping the rest of the methodology identical. Both follow the same evaluation protocol as Table 1. Results are reported in Table 5.

Semantic Conditioning (SGN-style). Following Zhang et al. [32], we augment each joint’s feature with an explicit joint-type embedding. Each of the J joint types is represented as a one-hot vector, passed through two fully connected layers (both 64-dimensional) to produce a learned embedding. This embedding is concatenated with the joint’s dynamics feature (a sum of separately encoded position and velocity, each also through two 64-dimensional FC layers). The resulting per-joint features are used to build a content-adaptive adjacency matrix via a non-local attention mechanism: two linear projections compute pairwise affinities across joints within each frame, followed by row-wise softmax normalization. Message passing proceeds through three residual GCN layers (dimensions 128, 256, 256). A spatial max-pooling step then aggregates all joints into a single frame-level vector, to which a learned frame-index embedding is added. Two temporal convolution layers (kernel sizes 3 and 1, dimensions 256 and 512) produce the final sequence representation. Batch normalization and ReLU are applied after each GCN and CNN layer.

Body-Part Typing (DS-GCN-style). Following Xie et al. [29], we decompose the spatial graph convolution into three parallel branches, each handling one-third of the feature channels. The skeleton is partitioned into five body parts (left/right arms, left/right legs, trunk with head), which defines both node types and edge types (an edge’s type is determined by the body-part labels of its two endpoint joints, giving 15 unique edge types). The *node type-aware branch* assigns separate projection kernels to each body part, so joints from different parts are mapped into distinct feature subspaces before computing pairwise affinities. The *edge type-aware branch* first computes a standard adaptive graph and then applies edge-type-specific transformations, so that connections between different body-part pairs (e.g., arm-to-trunk vs. within-trunk) are processed differently. The *general branch* uses a conventional adaptive graph with no semantic encoding. Each branch combines its adaptive graph with a fully learnable topology matrix via a branch-specific learnable weight. The three branch outputs are concatenated and fused through a 1×1 convolution. This structure is applied at every spatial GCN layer across all ten blocks of the network, with a base channel dimension of 64 (doubled at the 5th and 8th blocks). Training uses SGD with momentum 0.9, weight decay 5×10^{-4} , initial learning rate 0.1, and cosine annealing over 100 epochs.

B Training Details

We adapt UNVEIL to the Unitree G1 skeleton topology by constructing the adjacency matrix from the G1 kinematic tree. The model operates on input tensors of shape $(J \times C \times T)$, where $J = 15$ joints, $C=35$ channels (joint angles), $T=256$ frames (after downsampling from 120 fps to 30 fps). Task-specific output heads are attached to the trajectory embedding: a 522-class softmax for re-identification, a 2-class softmax for gender, and a single-output regression head with MSE loss for age, height, and weight. All experiments are conducted on a single NVIDIA RTX 4090 GPU (24 GB VRAM). Training a single task takes approximately 1300 seconds per epoch, totaling around 37 hours for a full 100-epoch run. Table 6 lists all hyperparameters.

Table 5: Operator attribute inference using two well-tested action-recognition architectures, SGN [32] and DSGCN [29], adapted to our pipeline. Full protocol matches Table 1.

	Task → Eval. Setting ↓	Gender Acc(%)↑	Weight MAE↓	Height MAE↓	Age MAE↓	Re-ID Top-1/5 Acc(%)↑	Linkage Acc(%)↑
SGN [32]	Unseen operators	78.2	10.9 kg	6.2 cm	5.5 yr	—	—
	Unseen demos of seen operators	81.4	8.9 kg	4.9 cm	4.1 yr	80.1/90.0	80.1
	<i>Performance range across 20 activity categories</i>						
	Most leaking activity	100.0	4.9 kg	4.5 cm	1.1 yr	100.0/100.0	100.0
	2nd-most leaking	92.0	5.9 kg	5.8 cm	2.2 yr	98.0/98.1	99.0
	2nd-least leaking	75.1	13.2 kg	9.1 cm	4.7 yr	69.6/85.1	73.1
	Least leaking activity	73.1	15.1 kg	11.9 cm	5.3 yr	68.5/83.2	71.2
	Unseen operators	79.1	10.1 kg	6.0 cm	4.8 yr	—	—
	Unseen demos of seen operators	82.8	4.9 kg	5.1 cm	3.9 yr	95.9/98.0	91.1
	<i>Performance range across 20 activity categories</i>						
DSGCN [29]	Most leaking activity	100.0	4.7 kg	4.1 cm	1.3 yr	100.0/100.0	100.0
	2nd-most leaking	93.3	5.5 kg	5.7 cm	2.6 yr	98.0/98.1	99.0
	2nd-least leaking	77.4	12.7 kg	9.3 cm	4.9 yr	70.1/88.4	72.6
	Least leaking activity	73.1	13.8 kg	12.2 cm	6.3 yr	72.1/82.7	71.2

C Emergent Behaviour Details

This appendix details the feature extraction, aggregation, and statistical procedures underlying the correlation and partial correlation analyses reported in Section 4.3.

Feature Extraction. Each motion demonstration is a Unitree G1 joint trajectory with 35 channels (6 root degrees of freedom and 29 joint angles) sampled at 120 Hz. From each demonstration we extract 27 interpretable features organized as nine body regions times three motion statistics.

We define body regions as: root translation, root rotation, hip, knee, ankle, waist, shoulder, elbow, and wrist. For bilaterally symmetric joints (hip, knee, ankle, shoulder, elbow, wrist), left and right channels are merged to reduce dimensionality while preserving coordination information. For each body region we compute three statistics: (1) *mean angular velocity*, the average magnitude of frame-to-frame angular differences scaled by the sampling rate; (2) *range of motion*, the difference between maximum and minimum joint angles over the demonstration; and (3) *peak angular velocity*, the maximum instantaneous angular velocity over the demonstration. For multi-axis regions (e.g., the three-DoF hip), all three statistics are aggregated across axes using the ℓ_2 norm. demonstrations shorter than 30 frames or containing fewer than 35 valid channels are discarded; remaining NaN and infinite values are zero-imputed.

Aggregation to the (Operator, Task) Level. Each operator typically performs multiple demonstrations per task. We compute the 27-feature vector per demonstration and then average across all demonstrations from the same operator performing the same task, yielding one feature vector per (operator, task) pair. The resulting data matrix is augmented with four attribute columns: `operator_age_yr`, `operator_height_cm`, `operator_weight_kg`, and `gender_numeric` (0 = female, 1 = male).

Task Filtering. Tasks are retained for analysis if they contain at least 10 demonstrations and 5 operators. For partial-correlation analysis we raise the operator threshold to 30 to ensure stable residual estimates. Headline results reported in the main text are restricted to the 18 tasks with at least 50 operators.

Raw Pearson Correlation. For each (task, feature, attribute) triple we compute Pearson’s r across all operators who performed that task. Pairs with fewer than five valid observations or near-zero variance in either variable are discarded. For the binary gender attribute we additionally report a tie-aware ROC AUC.

Partial Correlation. Raw correlations are susceptible to confounding because the four demographic attributes are mutually correlated in humans (taller operators weigh more; gender correlates with

Table 6: Structural and training hyperparameters for UNVEIL

Hyperparameter	Value
<i>Architecture</i>	
Input channels (C)	35
Sequence length (T)	256
Number of joints (J)	15
Native / target FPS	120 / 30
Number of ST blocks (N)	10
Base channels	64
Channel inflate stages	5, 8
Temporal downsample stages	5, 8
Channel ratio	2
Embedding dimension	256
Dropout	0.5
<i>Losses</i>	
Classification loss	Cross-entropy
Label smoothing	0.05
Regression loss	MSE
SupCon temperature (τ)	0.07
SupCon weight (λ)	0.1
SupCon warmup epochs	20
<i>Optimization</i>	
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-4}
Batch size	128
Max epochs	400
Gradient clip norm	5.0
Early stopping patience	40 evals
<i>Data Processing</i>	
Global normalization	zero mean, unit variance
Noise augmentation (σ)	0.01
Temporal masking fraction	0.05
Minimum sequence length	16

height; etc.). To isolate attribute-specific signal, we compute partial Pearson correlations that residualize each feature and target on the remaining three attributes. For each feature f and target attribute A , with control set $C = \{A_1, A_2, A_3\}$ denoting the other three attributes, we first residualize f on C by ordinary least squares ($f = X\beta_f + \varepsilon_f$, where $X = [\mathbf{1}, C]$), then residualize A on C analogously ($A = X\beta_A + \varepsilon_A$), and compute Pearson’s r between ε_f and ε_A to obtain the partial correlation r_{partial} . Intuitively, r_{partial} measures the feature–attribute association that remains after any linear combination of the other three attributes has been accounted for. A feature with high raw r but low r_{partial} was tracking confounders; a feature whose correlation is stable under partialing carries genuinely attribute-specific signal. Partial correlations are computed only when at least 30 operators remain after the validity mask, to ensure the residualization step does not dominate the statistic.

521 D Dataset Details

522 **Attribute Distributions.** We report the distribution of the four operator attributes in Figure 4. The
 523 dataset spans ages 17–71 yr, heights 145–199 cm, weights 38–145 kg, and has an approximately
 524 balanced gender split (241 F, 251 M).

Table 7: Mean raw and partial correlations for the most strongly correlated features per attribute, across tasks with ≥ 50 operators. Sign retention: k/n where n is the number of tasks in which the feature ranks among the top-10 correlated features for that attribute, and k is how many of those share the dominant correlation sign.

Attribute	Feature	Mean r	Mean r_{partial}	Sign Retention (top-10 tasks)
Age	waist_rom	-0.295	-0.250	14/14 (100%)
	hip_peak_vel	-0.294	-0.275	6/7 (85%)
	waist_mean_vel	-0.273	-0.233	12/12 (100%)
Height	root_translate_rom	-0.252	-0.246	9/11 (82%)
	root_translate_peak_vel	-0.191	-0.180	12/12 (100%)
	ankle_peak_vel	-0.162	-0.143	7/8 (88%)
Weight	root_translate_peak_vel	-0.247	-0.197	9/9 (100%)
	waist_rom	-0.203	-0.138	8/10 (80%)
	wrist_mean_vel	-0.151	-0.081	5/6 (83%)
Gender	ankle_rom	-0.312	-0.183	11/11 (100%)
	wrist_rom	-0.136	-0.126	11/11 (100%)
	shoulder_rom	-0.112	-0.10	10/10 (100%)

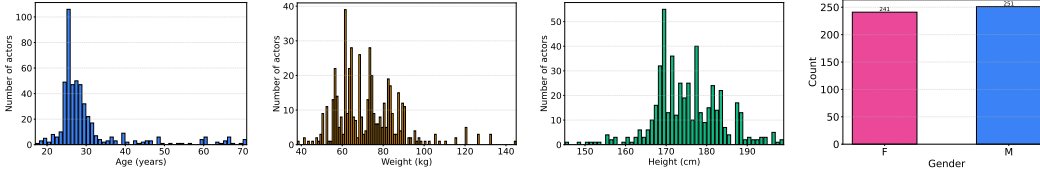


Figure 4: Distribution of user attributes in the dataset, including age, height, weight, and gender.

G1 Retargeted Trajectory Format. Each retargeted trajectory is a sequence of $J=15$ joint configurations with $C=35$ total channels, sampled at 120 Hz. Table 8 details the kinematic layout. The root provides 6 degrees of freedom (3 translational, 3 rotational); the remaining 29 channels are joint angles distributed across 12 actuated joints grouped into five kinematic chains: left leg (hip, knee, ankle), right leg (hip, knee, ankle), torso (waist), left arm (shoulder, elbow, wrist), and right arm (shoulder, elbow, wrist).

Table 8: Unitree G1 retargeted joint layout. Each row specifies a joint, its degrees of freedom, and the kinematic group it belongs to for UNVEIL’s graph construction.

Joint	Channels	DoF	Kinematic Group
Root translation	3	translateX/Y/Z	—
Root rotation	3	rotateX/Y/Z	—
Left hip	3	pitch, roll, yaw	Left leg
Left knee	1	pitch	Left leg
Left ankle	2	pitch, roll	Left leg
Right hip	3	pitch, roll, yaw	Right leg
Right knee	1	pitch	Right leg
Right ankle	2	pitch, roll	Right leg
Waist	3	pitch, roll, yaw	Torso
Left shoulder	3	pitch, roll, yaw	Left arm
Left elbow	1	pitch	Left arm
Left wrist	3	pitch, roll, yaw	Left arm
Right shoulder	3	pitch, roll, yaw	Right arm
Right elbow	1	pitch	Right arm
Right wrist	3	pitch, roll, yaw	Right arm
Total	35		

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